KP14 Cover's Theorem

> Level: undergraduate, year 3
>
> Pattern Recognition | Chapter 4, §4.13

Slides: read and download

Learning objectives

- Use Cover's formula to compute the number of linear dichotomies and the probability of separable for N points in general position in l dimensions.
- Explain the critical point $N = 2(l+1)$ and the phase transition, and predict the probability after lifting.
- Explain that lifting raises separability but does not guarantee generalization.

KP14 Cover's Theorem

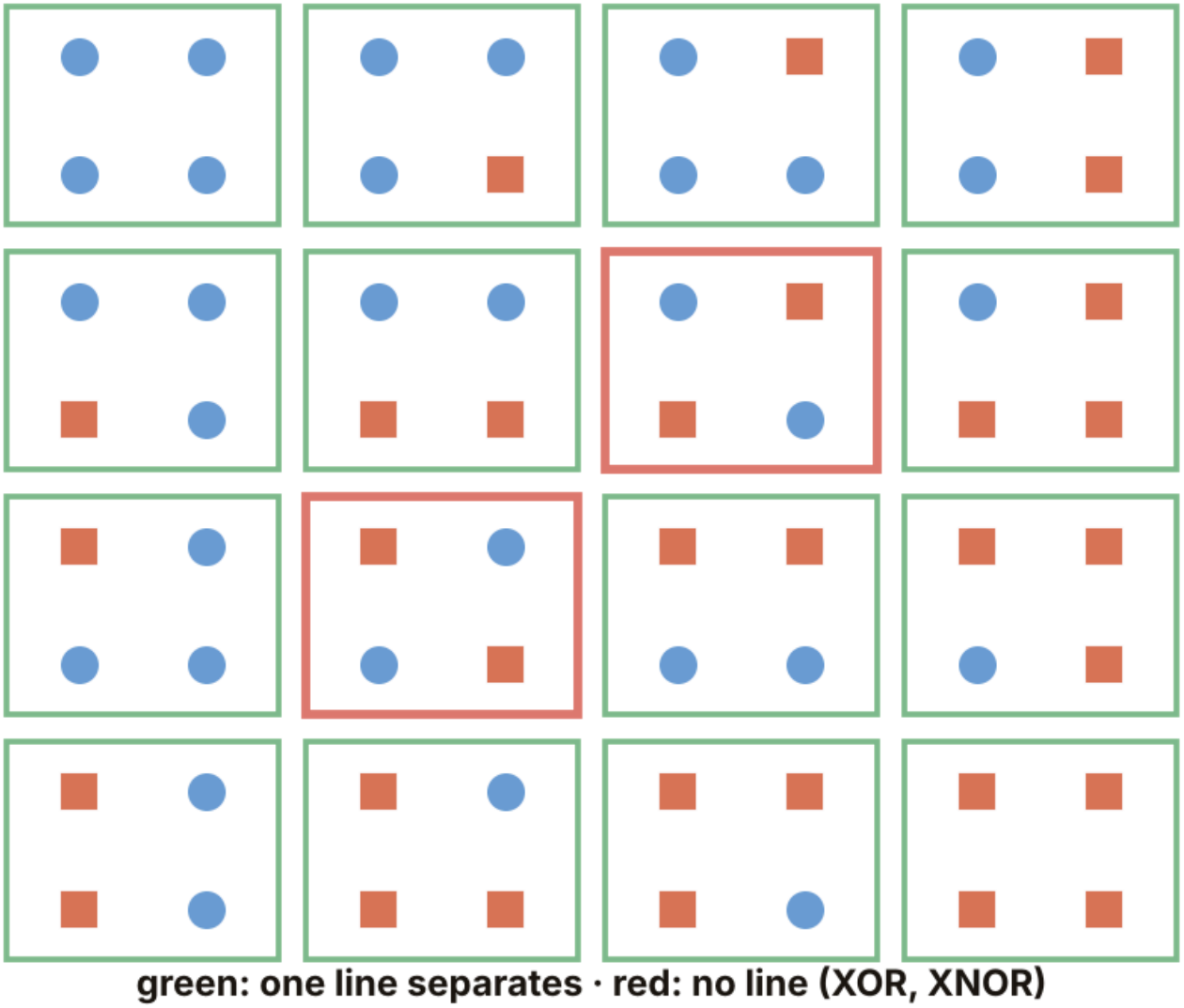
One-sentence summary: The higher the dimension, the more likely a set of points with random labels is still linearly separable, and there is a sharp critical point $N = 2(l+1)$; this is the quantitative licence for lifting, but it only guarantees that the training data can be separated, not that they generalize.

1. How many splits

Plain explanation: does lifting really help? First count into how many two-class splits one line can cut some points.

Dichotomy: one way of labeling N points with two classes; the ones a line (hyperplane) can realize are the linear dichotomies. For three points all 8 labelings are separable by a line; for four points at the corners of a square 14 of the 16 labelings are separable and the 2 that are not put the diagonal pairs together, $[(A, D), (B, C)]$, which are XOR and its mirror XNOR (A top-left, B bottom-left, C top-right, D bottom-right; checked one by one by a program).

Figure `lec05-kp14-dichotomy`: the 16 labelings, green separable and red not.



2. The counting formula

General position: points in l dimensions with no $l+1$ of them on one $(l-1)$-dimensional hyperplane (in two dimensions, no three on a line).

Formal definition (Cover's formula, source):

$$O(N,l) = 2\sum_{i=0}^{l}\binom{N-1}{i}.$$

Example: $N=4, l=2$: $2(1+3+3)=14$; $N=3$: $2(1+2+1)=8=2^3$. For $N \le l+1$, $O=2^N$: all labelings are separable. For $l=2$ and $N=1..8$, $O=2,4,8,14,22,32,44,58$ while $2^N=2,4,8,16,32,64,128,256$.

Figure `lec05-kp14-count`: $O(N,2)$ against 2^N.

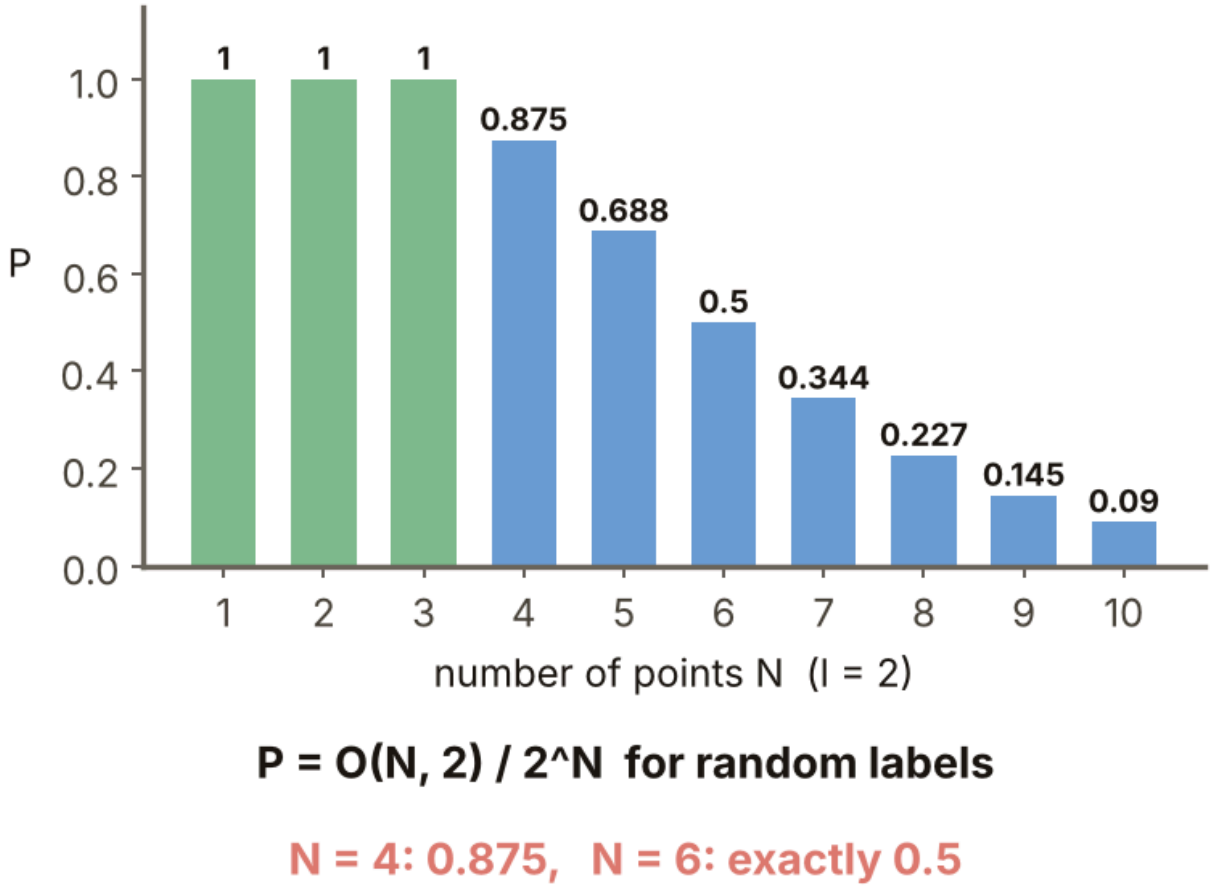
N	O(N, 2)	2^N (all labelings)
1	2	2
2	4	4
3	8	8
4	14	16
5	22	32
6	32	64
7	44	128
8	58	256

green: every labeling separable (N ≤ l + 1 = 3)

3. Probability of separable

With random labels, $P_N^l = O(N, l)/2^N$ and $P = 1$ for $N \leq l + 1$. In two dimensions ($l = 2$), for $N = 1..10$: $P = 1, 1, 1, 0.875, 0.6875, 0.5, 0.344, 0.227, 0.145, 0.090$. Once N passes $l + 1 = 3$ the probability starts to fall and it reaches exactly half at $N = 6$.

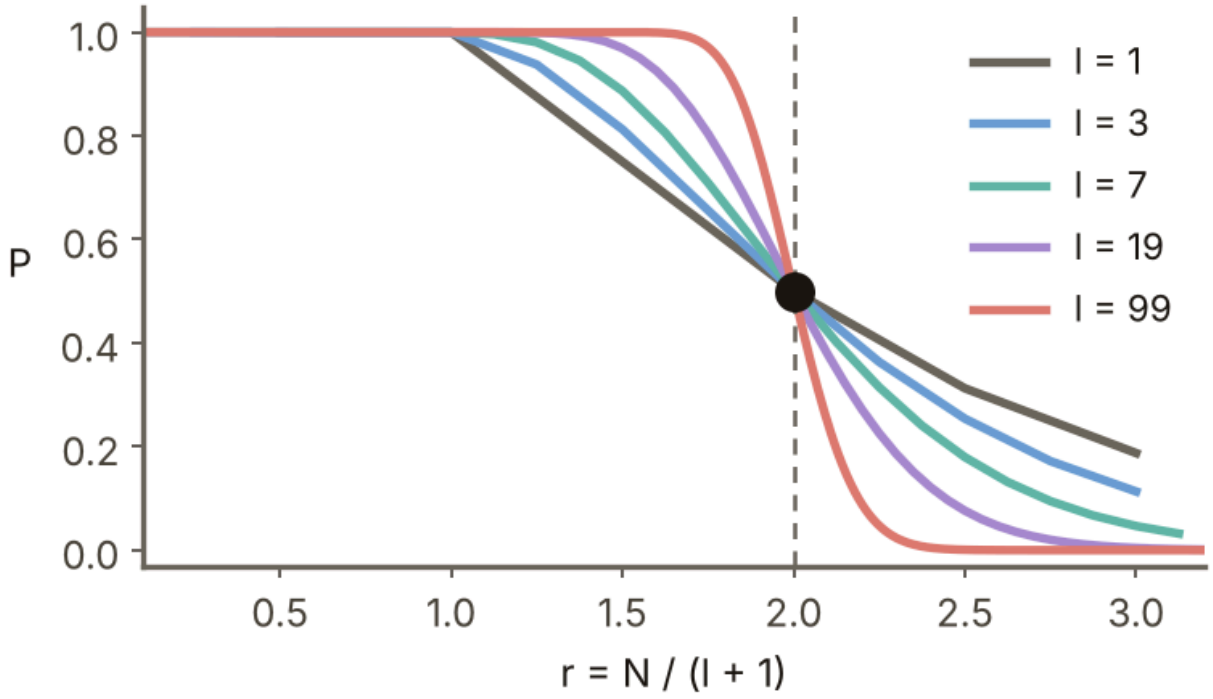
Figure `lec05-kp14-prob`: the probability for $l = 2$ and $N = 1..10$.



4. The phase transition

Phase transition: the turn of the probability from 1 down to 0 gets sharper as the dimension grows, like water freezing. With $N = r(l+1)$, at $r = 2$, i.e. $N = 2(l+1)$, $O(2l+2, l) = 2^{2l+1}$, so the probability is exactly $1/2$ (checked for $l = 1..7$). P at $r = 1.5$ and 2.5: $0.75, 0.31$ for $l = 1$; $0.97, 0.08$ for $l = 19$; $1.00, 0.0007$ for $l = 99$.

Figure `lec05-kp14-curves`: P against r for several l, all through $(2, 1/2)$.



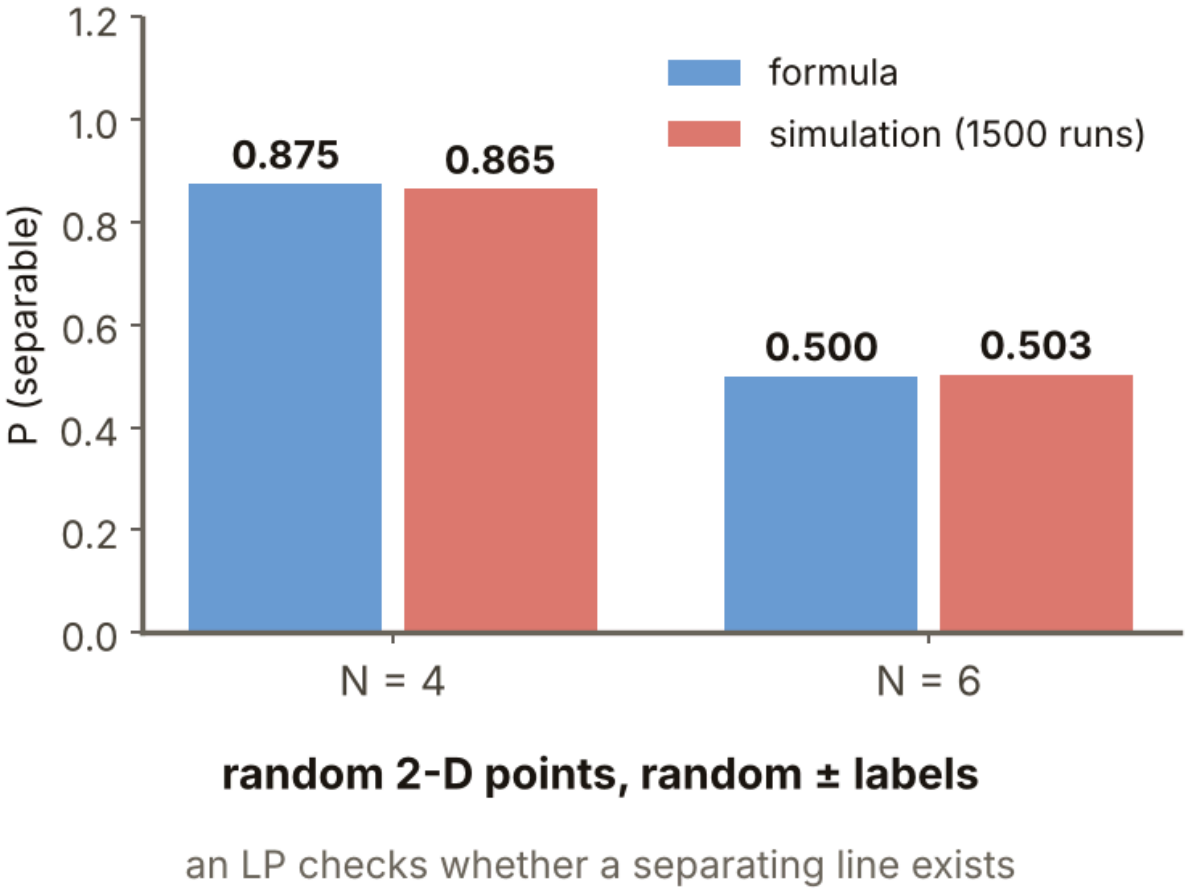
every curve passes through (r, P) = (2, 1/2)

r = 2.5: l = 1 → 0.31, l = 19 → 0.08, l = 99 → 0.0007

5. A random test

Each run draws N two-dimensional Gaussian points with random ± 1 labels and uses a linear program to decide whether a separating line exists, 1500 runs each: 0.865 at $N = 4$ (formula 0.875) and 0.503 at $N = 6$ (formula 0.500). With 1500 runs the standard error is about 0.01, so the gap is about one standard error (program, designed here).

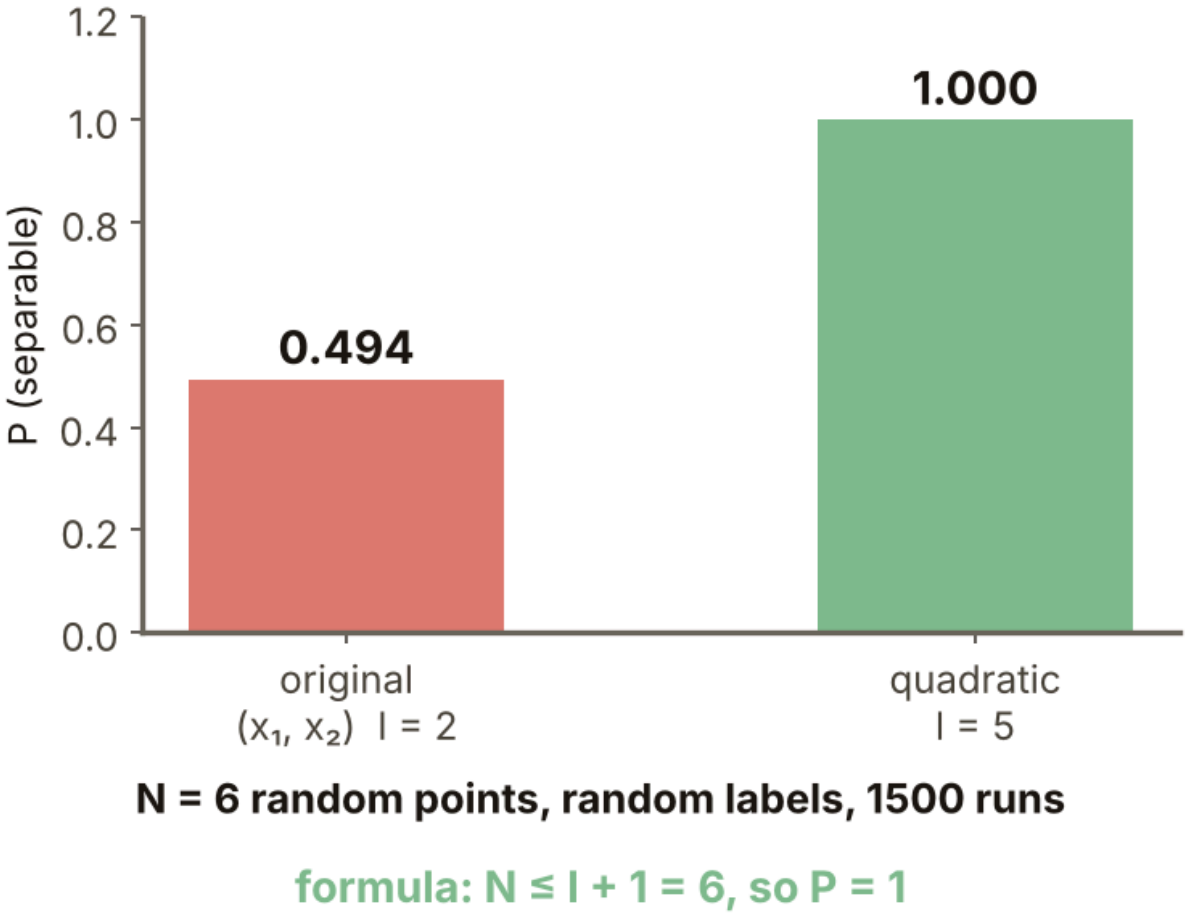
Figure `lec05-kp14-mc`: formula against simulation.



6. The effect of lifting

With the same random points and labels, $[x_1, x_2]$ is $l = 2$; adding $x_1^2, x_2^2, x_1 x_2$ makes $l = 5$. The formula predicts $N = 6 \leq l + 1 = 6$, so $P = 1$. The simulation (1500 runs) gives 0.494 for the original two dimensions and 1.000 for the quadratic features.

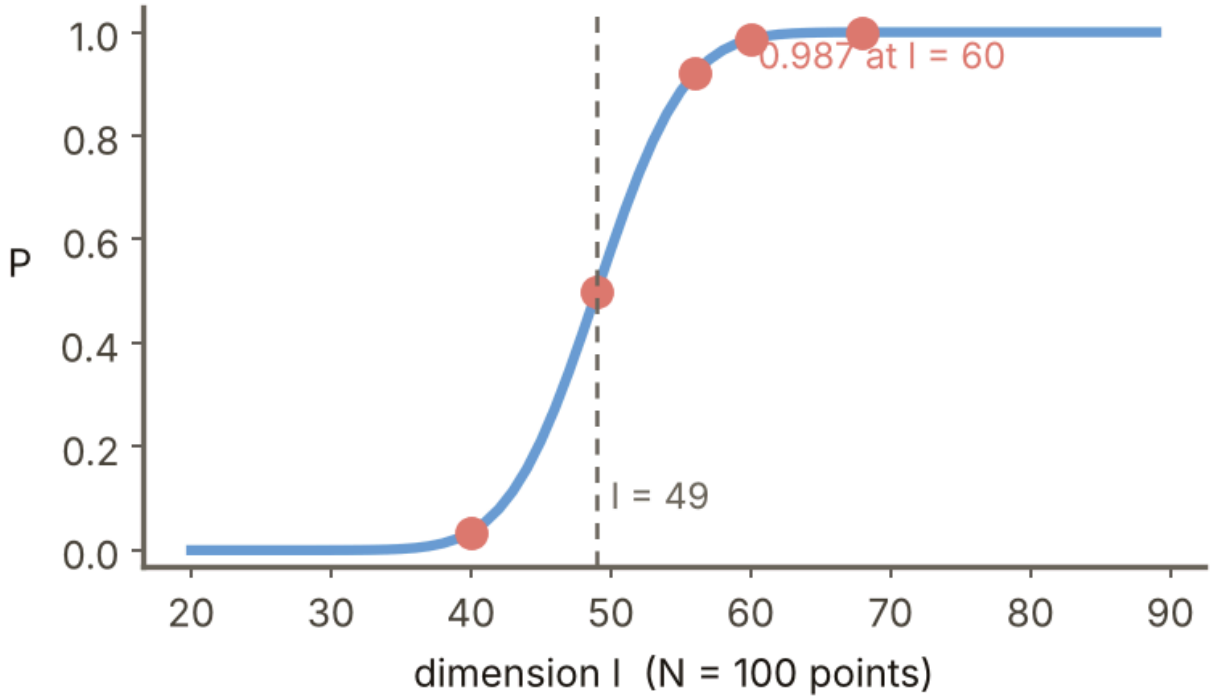
Figure `lec05-kp14-lift`: the separable fraction before and after lifting.



7. How many dimensions

For $N > l + 1$ some labelings fail; once the data exceed twice the degrees of freedom ($l + 1$ adjustable parameters), $N > 2(l + 1)$, the probability of separable drops below half. For $N = 100$: $l = 40$ gives 0.035; 48 gives 0.42; 49 exactly 0.5; 56 gives 0.92; 60 gives 0.987; 68 gives 0.9999. To separate random labels by lifting, the dimension must reach about half the number of samples or more.

Figure `lec05-kp14-need`: the probability for $N = 100$ against the dimension l.



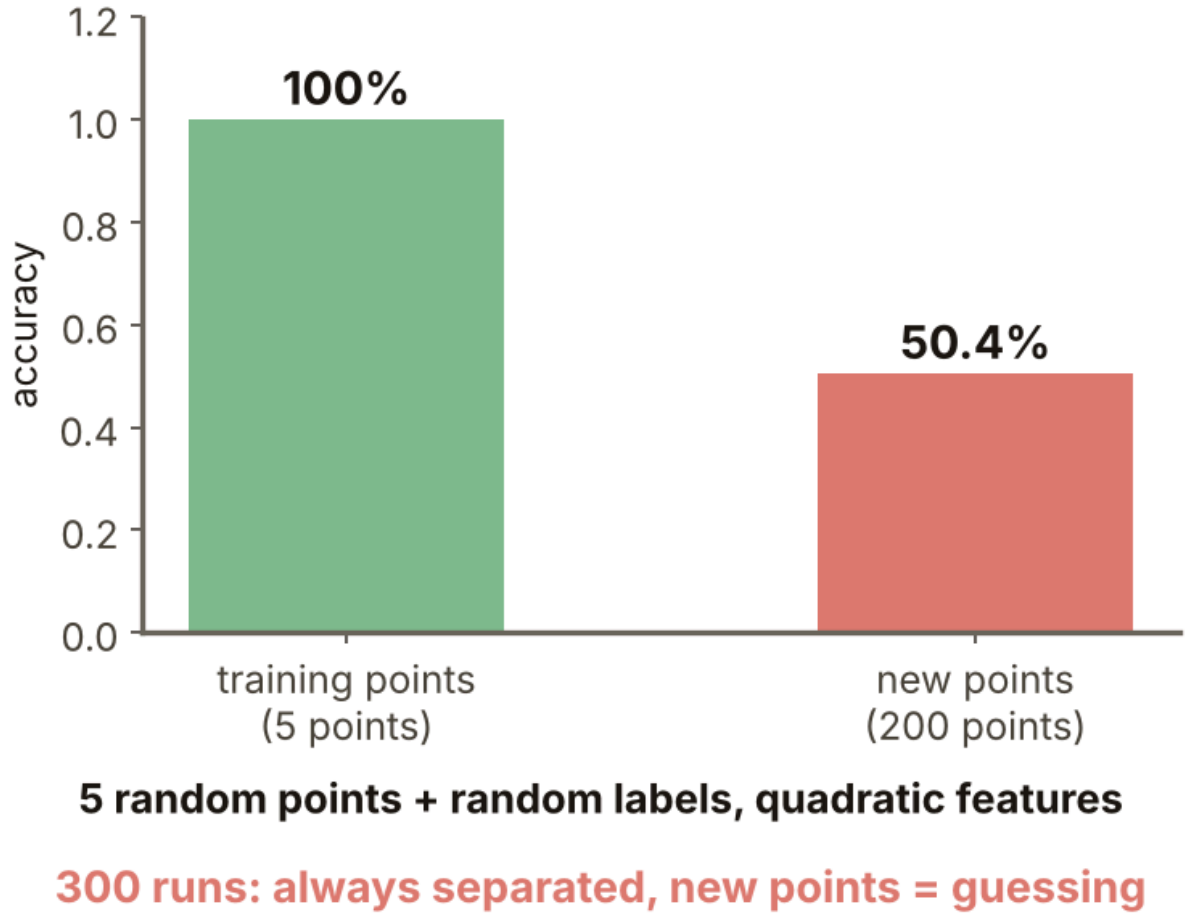
N = 100: I = 40 → 0.035, 48 → 0.42, 56 → 0.92

I = 49 is exactly 0.5 (N = 2(I + 1))

8. Separable is not general

Common misconception: after lifting, if the training data can be separated then new data will be classified correctly. Experiment (designed here): 5 two-dimensional Gaussian points with random labels and a separating hyperplane in the quadratic features (5 dimensions) was found in all 300 runs with training all right; then 200 fresh points, also with random labels, were only 50.4% right on average, which is guessing. Lifting only guarantees that the training data can be separated; only data with a pattern can be learned, and generalization is a separate matter (Chapter 5).

Figure `lec05-kp14-gen`: accuracy on training and new points.



Extension: why half, practice and worked solution

Why half: $\binom{2l+1}{i} = \binom{2l+1}{2l+1-i}$, so the sum over $i \le l$ is exactly half of the total 2^{2l+1}, namely 2^{2l}; $O(2l+2, l) = 2 \cdot 2^{2l} = 2^{2l+1}$ and $P = 1/2$. For $l = 1..5$, $O = 8, 32, 128, 512, 2048$.

Figure `lec05-kp14-proof`: O for $l = 1..5$ and the symmetry.

l	N = 2(l+1)	O(N, l) = 2^(2l+1)
1	4	8 = 2^3
2	6	32 = 2^5
3	8	128 = 2^7
4	10	512 = 2^9
5	12	2048 = 2^11

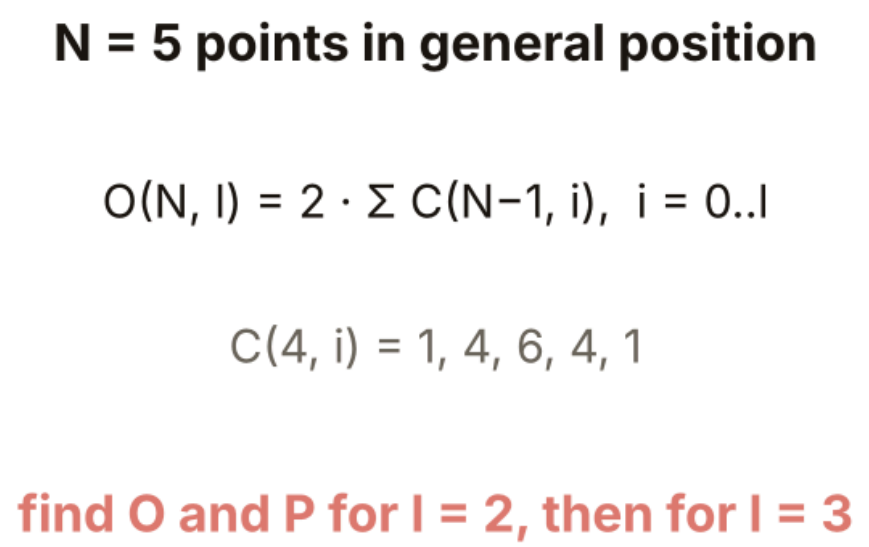
C(2l+1, i) = C(2l+1, 2l+1−i): the half with i ≤ l

is exactly half of 2^(2l+1), so P = 1/2

Practice: $N = 5$ points in general position.

1. What are $O(5, 2)$ and P for $l = 2$? Hint: $\binom{4}{i} = 1, 4, 6, 4, 1$.
2. Lifting to $l = 3$, what is P?

Figure `lec05-kp14-practice`: the setup of the practice.



Worked solution of questions 1 and 2: (1) $O(5,2) = 2(1+4+6) = 22$ and $P = 22/32 = 0.6875$. (2) $O(5,3) = 2(1+4+6+4) = 30$ and $P = 30/32 = 0.9375$. (3) Check: 22 matches the table of section 2, and one more dimension raises the probability of separable from 0.6875 to 0.9375.

Figure `lec05-kp14-practice-sol`: O and P for the two dimensions.

l = 2: 2(1 + 4 + 6)	= 22
P = 22 / 32	= 0.6875
l = 3: 2(1 + 4 + 6 + 4)	= 30
P = 30 / 32	= 0.9375

one more dimension: 0.6875 → 0.9375

Terms and sources

English	中文
Dichotomy	二分
General position	一般位置
Phase transition	相變

Source: Chapter 4 KP14, §4.13 (read 2026-10-25). Cover's formula, the probability of separable, the critical point $N = 2(l + 1)$ and the phase transition, 14 against 16 for $N = 4, l = 2$, and "lifting raises separability, not generalization" follow the source; every number, the random test (linear programming), the before-and-after lifting fractions, the $N = 100$ table and the random-label generalization experiment were designed and recomputed for this lecture ([added], (program)).